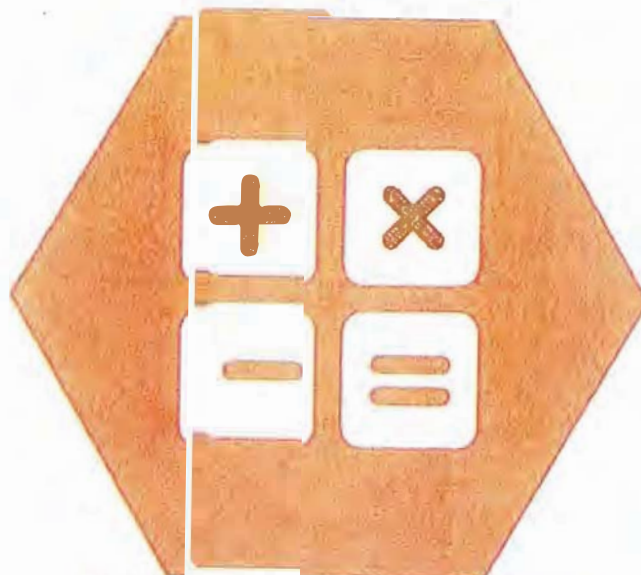




UNSW Global
AUSTRALIA

PAPER A



2016 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

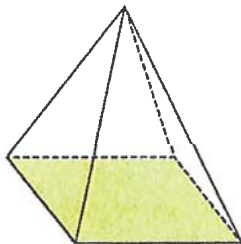
1. Today is Mike's birthday.



What age will Mike be on his birthday in 8 years' time?

- (A) 6
- (B) 8
- (C) 12
- (D) 14

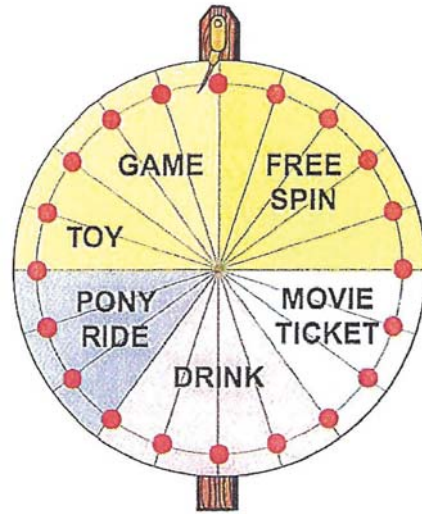
2. The base of this solid has 4 equal sides.



What shape is the base?

- (A) a square
- (B) a circle
- (C) a triangle
- (D) a pyramid

3. Mary spins this wheel.



What is the chance that Mary wins a toy?

- (A) certain
- (B) likely
- (C) unlikely
- (D) impossible

4. Which list gives three months, from the same year, in the correct order?

- (A) January, August, April
- (B) January, April, August
- (C) April, January, August
- (D) April, August, January

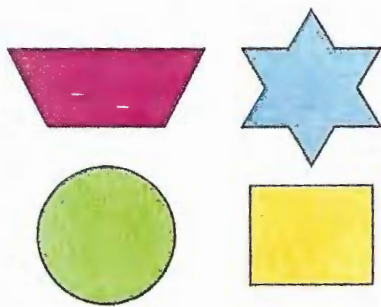
5. Jo wrote this number pattern.

1, 5, 9, 13, ?, 21, 25, ...

What number must ? be?

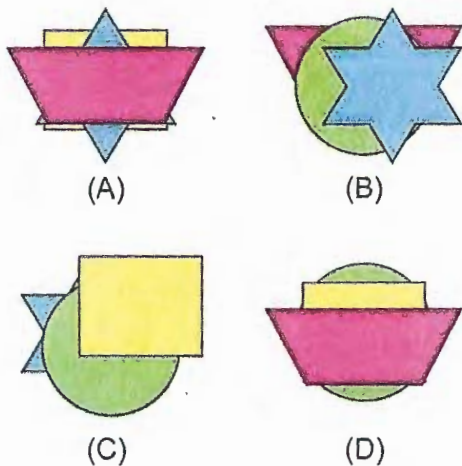
- (A) 14
- (B) 16
- (C) 17
- (D) 20

6. Jim had these four plastic shapes.



He put three shapes on top of each other.

In which of these is the rectangle on the bottom?



7. Which container will hold about one cup of liquid when full?



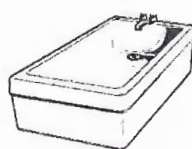
(A)



(B)



(C)



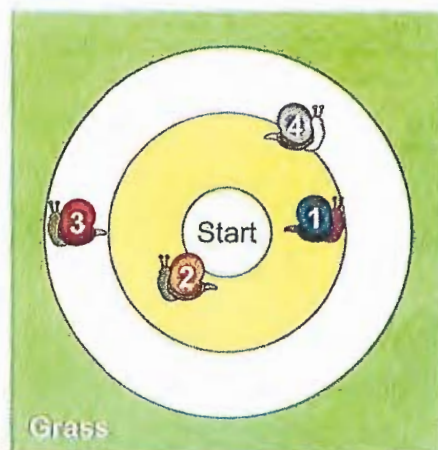
(D)

8. $24 - 9$ is _____ $8 + 9$

Which of these makes the statement true?

- (A) bigger than
- (B) less than
- (C) equal to
- (D) half of

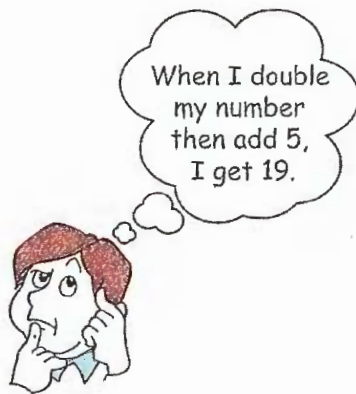
9. Four snails started in the centre circle. The picture shows their positions after 5 minutes.



Which list gives the order of the snails, from the closest to the grass to the furthest from the grass?

- (A) 3, 4, 1, 2
- (B) 3, 1, 4, 2
- (C) 1, 3, 4, 2
- (D) 1, 4, 3, 2

10. Danny thought of a number.

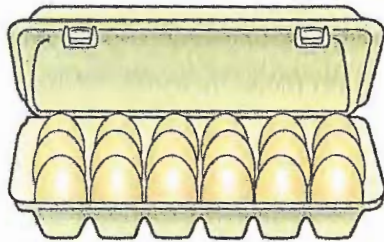


What was Danny's number?

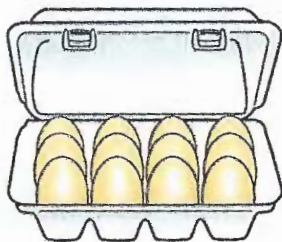
- (A) 7
- (B) 14
- (C) 43
- (D) 48

11. Jude has cartons in two sizes.

A large carton holds 18 eggs.



A small carton holds 12 eggs.



Jude filled 2 large cartons with eggs.

How many small cartons could Jude fill with these eggs?

- | | |
|-------|-------|
| (A) 6 | (B) 5 |
| (C) 4 | (D) 3 |

12. Sue drew a picture of a snail on her computer.



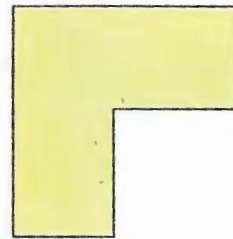
Next, Sue did something to the picture to make it look like this.



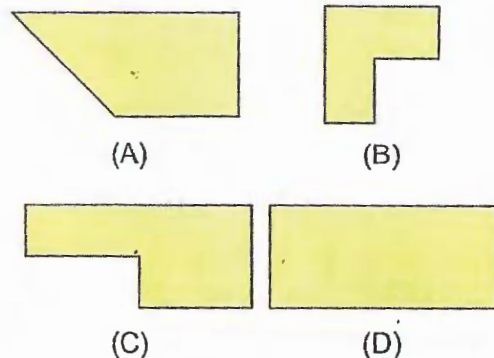
What did Sue do to the first picture to get the second picture?

- (A) She slid it.
- (B) She flipped it.
- (C) She turned it clockwise.
- (D) She turned it anticlockwise.

13. Ajay cut this shape in half.



What was the size and shape of each piece?



14. Lin circled her birthday on the calendar. She said, 'My birthday is 6 days from today!'

AUGUST 2015						
SUN	MON	TUE	WED	THU	FRI	SAT
30	31					1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

What was the date when Lin did this?

- (A) 11 August
(B) 12 August
(C) 17 August
(D) 23 August

15. $\boxed{?} - 7 = 35$

What must the value of $\boxed{?}$ be?

- (A) 28 (B) 32
(C) 38 (D) 42

16. All the children in Jed's class named their favourite sport.

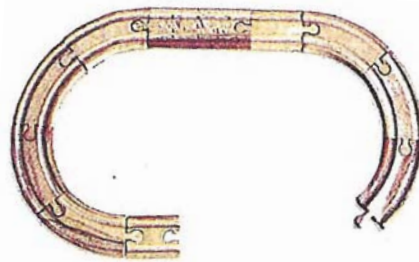
	Girls	Boys
Tennis	3	4
Soccer	7	8
Softball	$\boxed{?}$	5

There are two more boys in the class than girls.

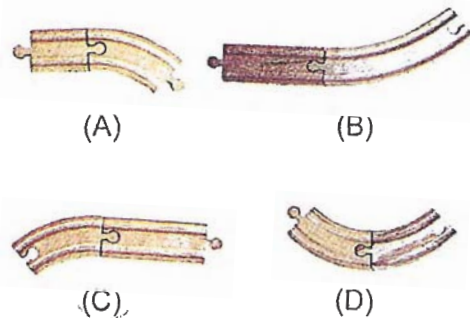
How many girls named softball as their favourite sport?

- (A) 4 (B) 5
(C) 6 (D) 9

17. Kim is putting a train track together.

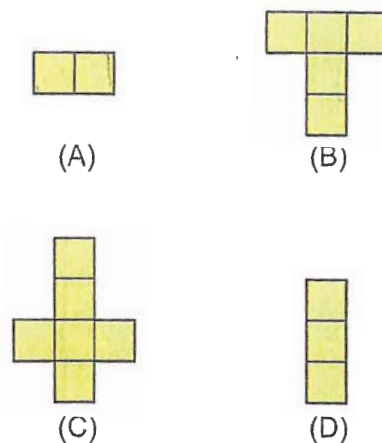


Which two pieces must she use to complete the track?



18. Jane is using squares to make some shapes. The side of each square is 1 unit long.

Which of these shapes has a distance of 6 units around the outside?



19. Pete made up this number pattern.

37	34	31	28	25	...
----	----	----	----	----	-----

What are the next three numbers in the pattern?

- (A)

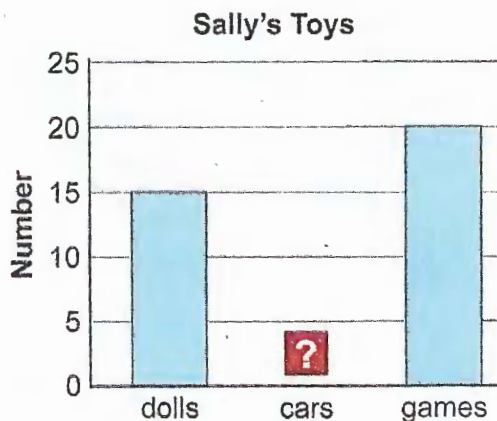
21	18	15
----	----	----
- (B)

22	17	14
----	----	----
- (C)

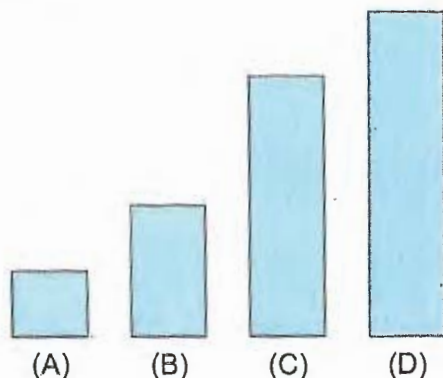
22	19	16
----	----	----
- (D)

24	23	22
----	----	----

20. Sally organised 45 of her toys into three groups.



Which of these should replace ? on the graph?



21. Mia's mother made a pizza. Mia shared the pizza with her two sisters.

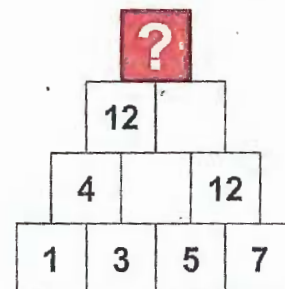


Mia's two sisters had equal amounts. Mia had as much as the two of them together. There was no pizza left.

How much of the pizza did Mia have?

- (A) half
- (B) one-third
- (C) one-quarter
- (D) three-quarters

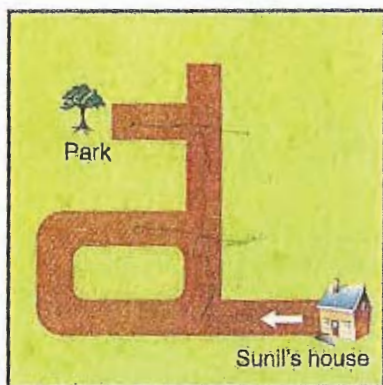
22. Sandy made a number pattern. She started on the bottom row. She added two numbers that were next to each other and wrote the answer in the box above. Some of the numbers are missing.



What must the value of ? be?

- (A) 22
- (B) 24
- (C) 32
- (D) 44

23. Sunil wants to walk along bush tracks from his house to the park.



Which route can he take?

- (A) the first track on the left and then the next track on the right
- (B) the first track on the left and then the second track on the right
- (C) the first track on the right and then the next track on the left
- (D) the first track on the right and then the second track on the left

24. Alex made a table to list some of the things he planned to do on Saturday.

Activity number	Time	Activity
1	8:00 – 9:00	walk the dog
2	9:30 – 11:30	play soccer
3	11:30 – 2:00	visit a friend
4	3:00 – 3:30	piano lesson

Which of these lists the activities in order from the shortest to the longest amount of time?

- (A) 1, 4, 2, 3
(B) 1, 4, 3, 2
(C) 4, 1, 2, 3
(D) 4, 1, 3, 2

25. Emma lives in Wonderland. In Wonderland one dollar equals 8 cents.

Emma bought a toy that cost 16
Wonderland dollars.

How much did the toy cost in Wonderland cents?

- (A) 2
(B) 14
(C) 24
(D) 128

26. Jane asked her friends, 'How much pocket money do you get each week?'

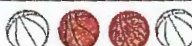
After they had all answered, Chris said, 'I get less than Abby but \$3 more than Deb.'

Billy said, 'I get \$1 less than Chris.'

Who gets the most pocket money?

- (A) Abby
(B) Billy
(C) Chris
(D) Deb

27. The graph shows the number of goals scored by the school basketball team over four weeks.

Week	Number of goals
1	
2	
3	
4	

 = 3 points

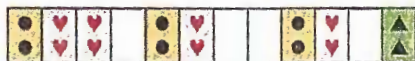
 = 2 points

 = 1 point

In which week did the team score the most points?

- (A) 1
(B) 2
(C) 3
(D) 4

28. Indra had 12 white cards. He drew designs on 8 of them.



Indra wanted the pattern to repeat after every 4 cards.

Which designs must he draw on the white cards?



(A)



(B)

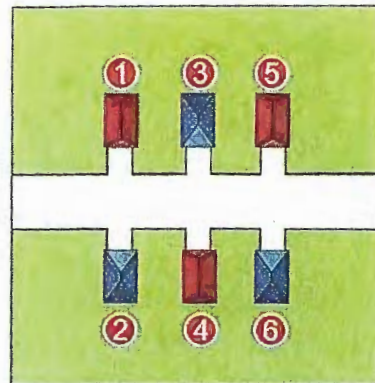


(C)



(D)

29. Colin, Aditi, Jake, Ben, Samira and Lin live in houses on the same street.



Samira lives in house number 5. Colin lives on the same side of the street as Samira.

Aditi lives in the house between Lin and Jake. Lin lives opposite Colin.

What is the number of the house that Ben lives in?

- (A) 4
(B) 3
(C) 2
(D) 1

30. Which of these objects is **NOT** a prism?



(A)



(B)



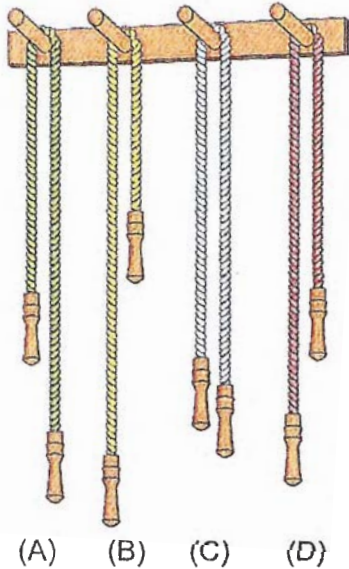
(C)



(D)

31. There are four skipping ropes hanging on hooks in the school hall.

Which skipping rope is the shortest?



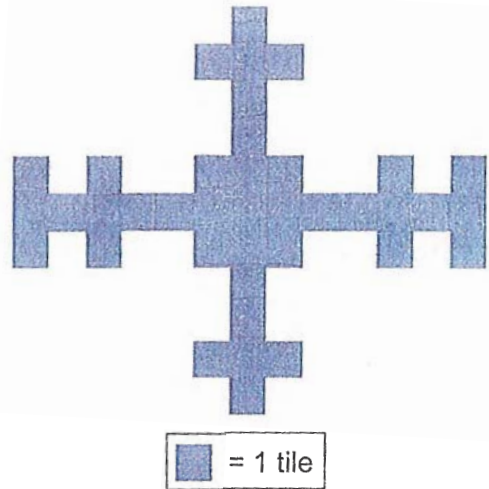
32. Lyna had a necklace with 32 coloured beads. One-quarter of the beads were yellow, one-eighth of the beads were blue and the rest were red.



How many of the beads were red?

- (A) 20
(B) 16
(C) 8
(D) 4

33. Aditi joined square tiles of the same size to make this shape. The tiles do not overlap.



How many tiles did she use?

- (A) 32
(B) 33
(C) 39
(D) 41

34. Ying lived on the 1st floor of a building. He walked up the stairs to visit his friend on the 5th floor.

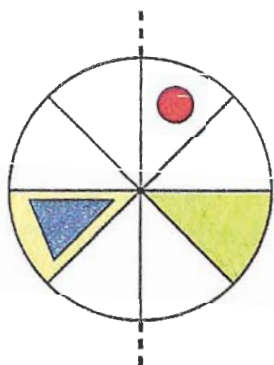
It took him 20 seconds to walk up to the 2nd floor and then another 25 seconds to walk up to the 3rd floor.

It took Ying 5 seconds longer each time he walked up to the next floor.

How long did it take for Ying to walk up the stairs to visit his friend?

- (A) 55 seconds
(B) 75 seconds
(C) 100 seconds
(D) 110 seconds

35. Jia made this glass pendant.

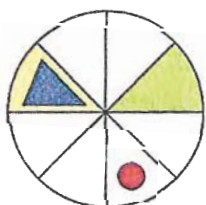


She flipped it over the dashed line and then rotated it anticlockwise by a quarter turn.

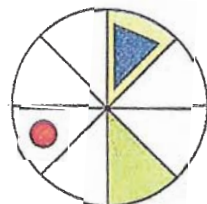
What does her pendant look like now?



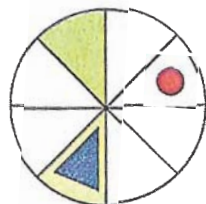
(A)



(B)



(C)



(D)

36. Jim's teacher asked the class to follow these steps:

- write down a two-digit number
- reverse the digits to make another two-digit number
- add the numbers together.

Sasi's number was 21.

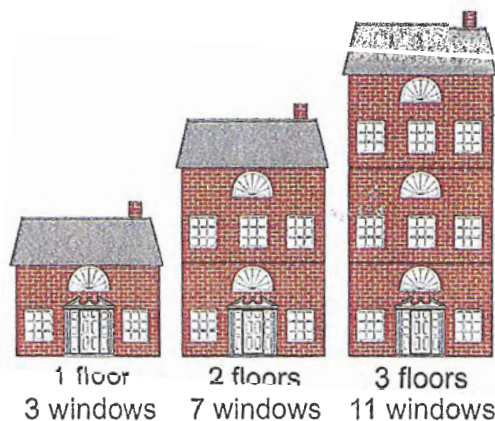
She correctly calculated $21 + 12 = 33$.

Jim's number was the smallest two-digit number that did **NOT** give a sum with all digits the same.

Which of these was Jim's number?

- (A) 19
(B) 17
(C) 15
(D) 10

37. Jessica is building some toy houses.



How many windows will a toy house with 6 floors have?

- (A) 19
(B) 22
(C) 23
(D) 27

38. Sven had to deliver 200 leaflets.
He looked at the clock as he left home.



Sven delivered 20 leaflets every
15 minutes.

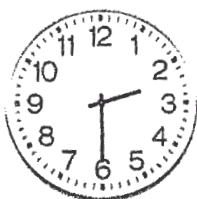
Which clock shows the time Sven finished
his delivery?



(A)



(B)



(C)



(D)

39. Danny has four bags of marbles.

Danny wants to pick a white marble
without looking.

From which bag must Danny take a
marble to have the greatest chance of
picking a white one?



(A)



(B)



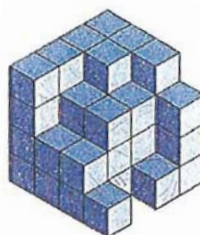
(C)



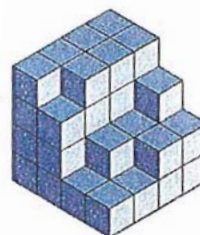
(D)

40. Lori used some small cubes to build a
large cube. She then took away 18 small
cubes from the large cube.

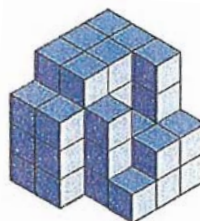
Which one of these shows Lori's
new solid?



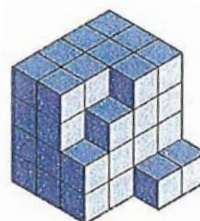
(A)



(B)



(C)



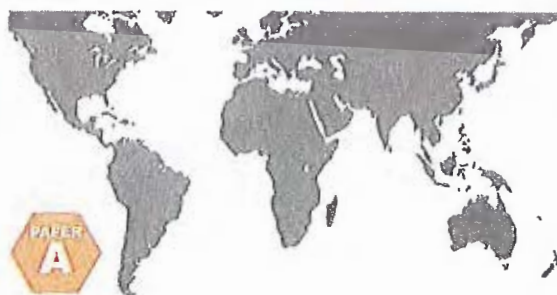
(D)

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The following year levels should sit THIS Paper:

Australia	Year 3
Brunel	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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IF THIS BOX APPEARS IN LEMON AND GREY TONE, THIS PAPER IS A JILLEGAL COPY

HOW TO FILL OUT THIS SHEET:



- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.
- Rub out all mistakes completely.

EXAMPLE 1: Debbie Bach

FIRST NAME

LAST NAME

EXAMPLE 2: Chan Ai Beng

FIRST NAME

LAST NAME

EXAMPLE 3: Jamal bin Abas

FIRST NAME

LAST NAME

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / /

Postcode: _____

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

CLASS
(optional)

	A	K
	S	L
	C	M
	D	N
	E	O
	F	P
	G	Q
	H	R
	I	S
	J	T

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
(B) 9
(C) 10
(D) 24

The answer is 10, so fill in the oval ☒, as shown.



START

- 1 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
2 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
3 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
4 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
5 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
6 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
7 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
8 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
9 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
10 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 11 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
12 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
13 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
14 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
15 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
16 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
17 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
18 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
19 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
20 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 21 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
22 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
23 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
24 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
25 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
26 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
27 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
28 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
29 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
30 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 31 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
32 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
33 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
34 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
35 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
36 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
37 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
38 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
39 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
40 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

ICAS Maths A 2016

Question Number	Answer
1	D
2	A
3	C
4	B
5	C
6	A
7	C
8	B
9	A
10	A
11	D
12	B
13	A
14	A
15	D
16	B
17	C
18	A
19	C
20	B
21	A
22	C
23	D
24	C
25	D
26	A
27	D
28	C
29	B
30	D
31	B
32	A
33	C
34	D
35	C
36	A
37	C
38	B
39	D
40	A